ECN 594: Two-Part Tariffs

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Plan

1. Two-part tariffs: structure and intuition
2. Optimal two-part tariff (homogeneous consumers)
3. Heterogeneous consumers (brief)

Non-linear pricing

- So far: price discrimination by **selection by indicators** and **self-selection**
  - Requires observable characteristics or menu of versions
- Now: **two-part tariffs**
  - Price per unit depends on quantity purchased
  - Consumers *self-select* into how much to buy
- Another form of non-linear pricing

Two-part tariff structure

- A two-part tariff has the form:

$$\text{Total payment} = F + p \times q$$

- F: fixed fee (membership, connection charge)
- p: per-unit price (price per unit consumed)
- q: quantity consumed
- **Examples:**
  - Golf club: membership fee + greens fee per round
  - Phone plan: monthly fee + per-minute charges
  - Costco: annual membership + price per item

Why is this “non-linear”?

- Average price per unit: $\frac{F+pq}{q} = p + \frac{F}{q}$
- As q increases, average price **decreases**
- This is a quantity discount
- The pricing is “non-linear” because total payment is not proportional to quantity

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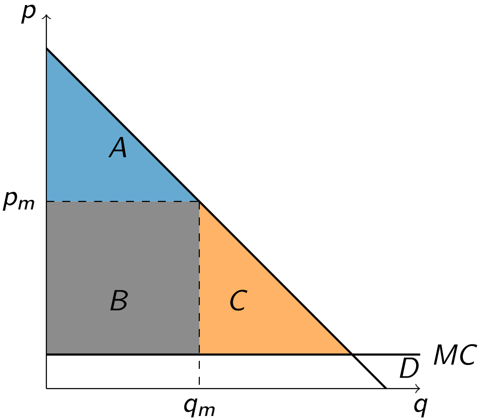
Why use two-part tariffs?



- Under uniform pricing at p_m:
  - CS $=$ Area A
  - Profit $=$ Area B
  - DWL $=$ Area C
- Monopolist "leaves money on the table"
- Can we do better?

Optimal two-part tariff: homogeneous consumers

- Assume all consumers are identical (same demand curve)
- **Key insight:** Use F to extract surplus, use p to maximize total surplus
- **Step 1:** Set $p = MC$
  - This maximizes total surplus (eliminates DWL)
- **Step 2:** Set $F = CS(p = MC)$
  - Extract all consumer surplus through the fixed fee

Why price at marginal cost?

- Firm profit $= CS(p) + (p - c)q(p) =$ Total surplus
- Total surplus is maximized at $p = c$ (no deadweight loss)
- **Intuition:**
  - High p: Earn margin on sales, but reduce CS (and hence F)
  - Low p: Lose margin, but CS increases more
  - Optimal: $p = c$, extract all surplus via F
- **Result:** $p^* = c$, $F^* = CS(c)$, $\pi^* = CS(c)$

Deriving $p^* = c$ with calculus

- For linear demand $q = a - bp$:

$$CS(p) = \frac{(a - bp)^2}{2b}$$

- Profit: $\pi = CS(p) + (p - c)(a - bp)$

- **FOC:**

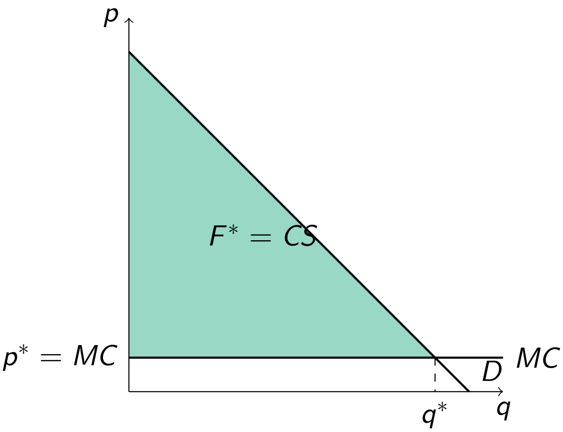
$$\frac{d\pi}{dp} = \frac{dCS}{dp} + (a - bp) - b(p - c) = 0$$

- Since $\frac{dCS}{dp} = -(a - bp)$:

$$-(a - bp) + (a - bp) - b(p - c) = 0$$

- Simplifies to: $p^* = c$

Optimal two-part tariff: graphically



- Set $p^* = MC$
- Consumers buy q^* (efficient quantity)
- Set F^* = entire green triangle
- Profit $= F^* - 0 = F^*$
- **Result:** Firm captures ALL surplus

Worked example: Sushi bar

- **Question:** A sushi bar has $MC = 0.1$ per piece. Each consumer has demand $q = 20 - 10p$.
- (a) What is the optimal uniform price per piece and profit per consumer?
- (b) Consider an all-you-can-eat policy. What is the optimal fee and profit per consumer?
- (c) What is the optimal two-part tariff (fee at the door plus price per piece)?

Worked example: Sushi bar (solution)

- **(a) Uniform pricing:**
  - Inverse demand: $p = 2 - 0.1q$
  - Profit: $\pi = (p - 0.1)q = (1.9 - 0.1q)q = 1.9q - 0.1q^2$
  - FOC: $\frac{d\pi}{dq} = 1.9 - 0.2q = 0 \Rightarrow q = 9.5$
  - Price: $p = 2 - 0.1(9.5) = 1.05$
  - Profit: $\pi = (1.05 - 0.1) \times 9.5 = 9.025$
- **(b) All-you-can-eat** ($p = 0$, charge fee F):
  - At $p = 0$: $q = 20$, $CS = \frac{1}{2} \times 20 \times 2 = 20$
  - Set $F = 20$. Cost $= 0.1 \times 20 = 2$
  - Profit: $\pi = 20 - 2 = 18$ (much higher)

Worked example: Sushi bar (solution continued)

- **(c) Optimal two-part tariff:**
  - Set $p = MC = 0.1$
  - At $p = 0.1$: $q = 20 - 10(0.1) = 19$
  - $CS = \frac{1}{2} \times 19 \times (2 - 0.1) = \frac{1}{2} \times 19 \times 1.9 = 18.05$
  - Set $F = 18.05$
  - Profit from sales: $(0.1 - 0.1) \times 19 = 0$
  - Total profit: $\pi = 18.05$
- **Summary:**
  - Uniform pricing: $\pi = 9.025$
  - All-you-can-eat: $\pi = 18$
  - Two-part tariff ($p = MC$): $\pi = 18.05$ (best)

Welfare effects of two-part tariffs

- Compared to uniform monopoly pricing:
- **Producer surplus:** Increases (captures all surplus)
- **Consumer surplus:** Decreases to zero
- **Total surplus:** Increases (DWL eliminated)
- **Efficiency vs equity tradeoff:**
    - More efficient (no DWL)
    - Less equitable (consumers get nothing)

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What about heterogeneous consumers?

- Reality: consumers have different demand curves
- Problem: if F is too high, some consumers don't participate
- **Tradeoff:**
  - High F: extract more from high-value consumers, but lose low-value ones
  - Low F: serve more consumers, but extract less per consumer
- Optimal F balances these effects

Worked example: Two-part tariff with two types

- **Question:**
- Type H: demand $q_H = 10 - p$, 100 consumers
- Type L: demand $q_L = 5 - p$, 200 consumers
- $MC = 0$
- Compare: (a) $F = 50, p = 0$ vs (b) $F = 12.5, p = 0$

Worked example: Two-part tariff with two types (solution)

- **(a)** $F = 50, p = 0$**:**
  - Type H: $CS = \frac{1}{2}(10)(10) = 50 \geq F$, joins
  - Type L: $CS = \frac{1}{2}(5)(5) = 12.5 < F$ doesn't join
  - Profit $= 50 \times 100 = \$5,000$
- **(b)** $F = 12.5, p = 0$**:**
  - Type H: $CS = 50 \geq 12.5$, joins
  - Type L: $CS = 12.5 \geq 12.5$, joins
  - Profit $= 12.5 \times 300 = \$3,750$
- Better to serve only H (in this case)

Key Points

1. **Two-part tariff:** $F + p \times q$ (fixed fee + per-unit price)
2. **Optimal (homogeneous):** Set $p = MC$, $F = CS$ at that price
3. Two-part tariffs increase efficiency but reduce consumer surplus to zero
4. With heterogeneous consumers: tradeoff between extraction and participation